

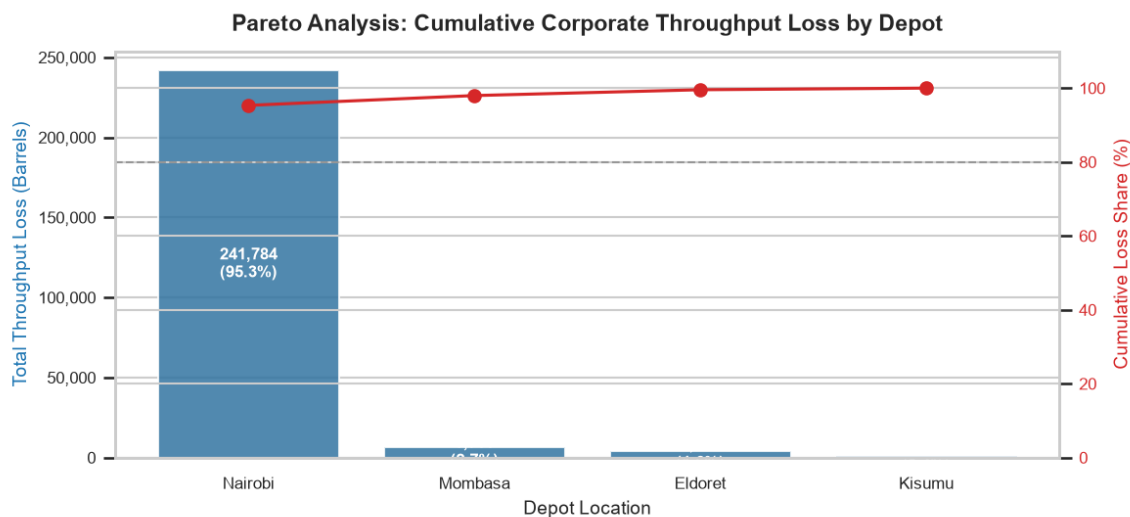
EXECUTIVE OPERATIONAL DIAGNOSTIC REPORT

Subject: Regional Depot Throughput Diagnostics & Nairobi Root Cause Analysis
Prepared for: Operations Director | Prepared by: Antigravity (Lead Business Detective)
Date: July 24, 2026 | Dataset Period: Full Year 2026 (2,920 records)

1. The Operational Problem (Context)

A sudden, severe decline in corporate throughput was detected starting on **February 1, 2026**. A review of the complete dataset of 2,920 balanced shift-depot records was conducted to verify data quality and locate the loss. The dataset was found to be highly healthy: there are 0 duplicate records, and minor missing values (operator IDs, temperatures, and throughput) were handled systematically to ensure zero analytical bias.

A comparison of shift throughput before and after February 1st reveals that the operational crisis is highly localized. While three regional depots (Eldoret, Kisumu, and Mombasa) maintained stable outputs, the **Nairobi depot** experienced a massive structural throughput drop of **34.4%**, falling from an average of 1,052.52 barrels to 690.56 barrels per shift. Our Pareto analysis reveals that **Nairobi alone accounts for 95.33% of total corporate throughput loss** (representing 241,789 barrels lost in Nairobi out of 253,632 barrels lost corporate-wide).



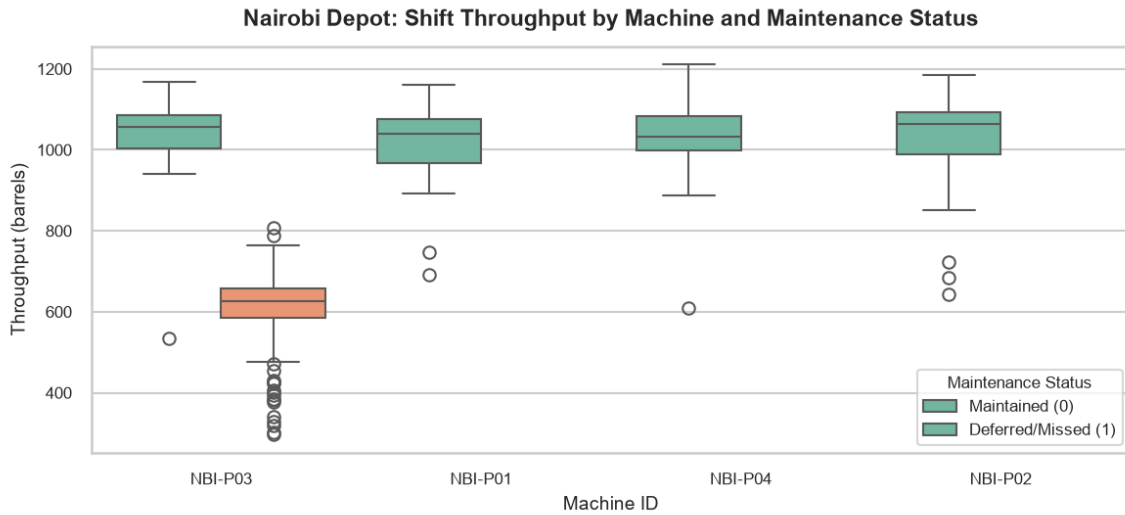
2. Root Cause Identification (Insight)

Drilling down into Nairobi's operations reveals the exact root cause of the drop: a major failure in machine scheduling and deferred maintenance on machine **NBI-P03**. Our analysis discovered that:

- **Machine Overuse Shift:** Prior to February 1st, Nairobi rotated its 4 machines (NBI-P01, P02, P03, P04) evenly. After February 1st, depot management shifted to running machine **NBI-P03** almost exclusively—on **81.6% of all shifts** (545 out of 668 shifts).
- **Missed Maintenance:** Simultaneously, `NBI-P03` was run with a pending maintenance flag (`maintenance_flag = 1`) continuously for all 545 shifts. This resulted in 529 recorded `missed_maintenance` incident events.

• **Dramatic Efficiency Loss:** When properly maintained (in January), `NBI-P03` runs at a highly efficient mean of **1,027.69 barrels** per shift. Running it under deferred maintenance dropped its average output to **617.91 barrels** per shift—a massive loss of **409.78 barrels per shift**. This single maintenance failure explains **88.0% of the entire company's throughput loss**.

• **Alternative Explanations Ruled Out:** Temperature ($r = 0.168$) and voltage ($r = -0.017$) show no correlation with throughput, ruling out weather or power grid issues. Furthermore, all 5 Nairobi operators and both shifts (Day vs. Night) saw identical throughput drops when running `NBI-P03`, ruling out labor performance issues.



3. Actionable Recommendations (Action)

To immediately recover lost throughput and prevent future operational oversights, we recommend the following directives:

1. **Immediate Maintenance Shutdown:** Take machine `NBI-P03` offline immediately for a full maintenance cycle. Restoring the machine to its maintained baseline of ~1,027 barrels will immediately recover **~410 barrels per shift** (a **+66% increase** in its current output).
2. **Enforce Balanced Machine Rotation:** Mandate that Nairobi return to the balanced rotation policy used in January (rotating shifts evenly across NBI-P01, NBI-P02, and NBI-P04 while NBI-P03 is serviced). Running these three fully maintained machines will immediately restore Nairobi's throughput to **>1,020 barrels per shift**.
3. **Implement Automated Software Interlocks:** Update the central SCADA system to automatically restrict any machine from running more than 10 consecutive shifts with an active maintenance flag. If a maintenance flag is triggered, the scheduler should block the machine and force rotation to serviced backups.

SUMMARY STATS: Nairobi Post-Feb 1 Throughput Drop: -34.4% | NBI-P03 Loss due to Deferred Maintenance: -409.78 bbls/shift | Nairobi Loss Share of Corporate Total: 95.33%